

Week 6 Assignment

Total marks=20

Q1. At 25°C, the specific conductance of a solution containing 3.20 g of BaCl₂ per litre is $4.10 \times 10^{-3} \text{ mho cm}^{-1}$. The equivalent conductance of the solution is: (Molar mass BaCl₂ = 208 g/mol, Eq wt = 104 g/eq)

- a. 133 mho cm² Normality = $\frac{\text{Molar}}{\text{Eq weight}} \times \text{volume} = 11 \times \frac{3.20}{104 \times 1} = 0.0307 \text{ N/l}$
b. 1330 mho cm²
c. 410 mho cm²
d. 82.5 mho cm²

$$\lambda_{eq} = \frac{\kappa \times 1000}{N} = \lambda_{eq} = \frac{4.10 \times 10^{-3} \times 1000}{0.0307} = \frac{41.0}{0.0307} \approx 133.25 \text{ mho cm}^2 \text{ eq}^{-1} \quad (2)$$

Q2. On diluting a strong electrolyte, the equivalent conductance:

- a. Remains constant
b. Decreases
c. First decreases then increases
d. Increases

(2)

Q3. The equivalent conductance at infinite dilution for NaCl, KCl, KNO₃, and HNO₃, at 300K are 128, 150, 162, and 425 respectively. Hence λ_0 for HCl in the same unit will be

- a. 451
b. 413
c. 387
d. 425

$$\lambda_{HCl}^0 = \lambda_{HNO_3}^0 + \lambda_{KCl}^0 - \lambda_{KNO_3}^0$$
$$\lambda_{HCl}^0 = 425 + 150 - 162$$
$$\lambda_{HCl}^0 = 575 - 162 = 413$$

(2)

Q4. Which expression(s) can be used to calculate $\lambda_0(\text{H}_2\text{O})$?

- A. $\lambda_0(\text{HCl}) + \lambda_0(\text{KOH}) - \lambda_0(\text{KCl})$
B. $\lambda_0(\text{HNO}_3) + \lambda_0(\text{NaOH}) - \lambda_0(\text{NaNO}_3)$
C. $\lambda_0(\text{NH}_4\text{OH}) + \lambda_0(\text{HCl}) - \lambda_0(\text{NH}_4\text{Cl})$
D. $\lambda_0(\text{H}_2\text{SO}_4) - \lambda_0(\text{HSO}_4^-)$

Correct option:

- a. A, C
- b. Only A
- c. A, B, C
- d. Only C

(2)

Q5. If the concentration is increased to 4 times its original value, the change in equivalent conductance for strong electrolytes is (where b is Kohlrausch's constant)

- a. 0
- b. $b\sqrt{c}$
- c. $2b\sqrt{c}$
- d. $4b\sqrt{c}$

(2)

Q6. At 25 °C the ionic mobilities of Mg^{2+} and SO_4^{2-} are $5.3 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $5.1 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively. What is the equivalent conductance of MgSO_4 ?

- a. 100.4 mho cm^2
- b. 108.0 mho cm^2
- c. 1.04×10^{-3} mho cm^2
- d. 53.0 mho cm^2

$$\begin{aligned} \lambda_{eq} &= F[u_+ + u_-] \\ \lambda_{eq} &= 96500 \times (5.3 \times 10^{-4} + 5.1 \times 10^{-4}) \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1} \\ \lambda_{eq} &= 96500 \times (10.4 \times 10^{-4}) \\ \lambda_{eq} &= 96500 \times 0.00104 = 100.36 \text{ mho} \cdot \text{cm}^2 \end{aligned}$$

(2)

Q6. A conductivity titration curve shows a sharp decrease in conductance initially, and a sharp increase after the equivalence point. This corresponds to:

- a) Strong acid vs strong base
- b) Strong acid vs weak base
- c) Weak acid vs weak base
- d) Weak acid vs strong base

(2)

Q7. Determine the ionic strength for a mixture of BaCl_2 , and KCl solution if $m_\pm = 0.008$.

2

- a) 0.028
- b) 0.032
- c) 0.04
- d) 0.056